

Applying all model configurations in one line in Entity Framework Core

When working with EF Core, you often have entity configuration classes that implement `IEntityTypeConfiguration<TEntity>`. Instead of registering each configuration manually, EF Core provides a mechanism to automatically apply all configurations in one step.

The main concept here is:

- **Assembly Scanning:** EF Core can scan your assembly for all types implementing `IEntityTypeConfiguration<T>` and apply them automatically.
- This eliminates the need to call `modelBuilder.ApplyConfiguration(...)` for each entity.

So in short, the one-line way to apply all configurations is by telling EF Core to apply configurations from an assembly. This keeps your `OnModelCreating` method clean and ensures all entity configurations are picked up consistently.

"Up to new Change" vs "Up to previous Change" in EF Core migrations

These concepts relate to migrating the database schema forward or backward.

Up to new Change (migrating forward)

- Means applying the latest migration that you created.
- When you add a migration, EF Core generates schema changes as C# code.
- Running the update "up to new change" applies all pending migrations in chronological order until the database is fully up-to-date with your code model.
- Think of it as bringing the database schema forward to match the current model definition.

Up to previous Change (rolling back)

- Means reverting the database schema back to the state before the last migration (or a chosen one).
- Instead of applying the latest changes, you're undoing them.
- This is useful if you introduced a breaking schema change or if you want to test a rollback scenario.
- Conceptually, you are taking the database backward in time, to a state that matched your code model before the last migration was applied.